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# The Influence of Injury History on Countermovement Jump Performance and Movement Strategy in Professional Soccer Players: Implications for Profiling and Rehabilitation Foci

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***Context:*** Professional soccer players who have sustained a lower limb injury are up to 3× more likely to suffer a reinjury, often of increased severity. Previous injury has been shown to induce compensatory strategies during neuromuscular screening tests, which might mask deficits and lead to misinterpretation of readiness to play based on task outcome measures. ***Objective:*** To investigate the influence of previous injury in professional soccer players on countermovement jump (CMJ) performance and movement strategy. ***Design:*** Cross-sectional. ***Setting:*** Professional soccer club competing in the English Championship (tier 2). ***Patients (or Other Participants):*** Outfield players with a minimum of 6 years as a professional. ***Intervention(s):*** Players were categorized as previously injured (n = 10) or not injured (n = 10). All players completed double- and single-leg CMJ trials. ***Main Outcome Measures:*** CMJ performance was quantified as jump height and flight time:contraction time ratio. CMJ movement strategy was quantified as force–time history, differentiating eccentric and concentric phases and CMJ depth. ***Results:*** Double-leg CMJ was not sensitive to previous injury in performance or movement strategy. In contrast, single-leg CMJ performance was impaired in players with previous injury, who generated significantly lower eccentric and concentric peak force and rate of force development, and a deeper countermovement. Impaired single-leg CMJ performance was also evident in the nonaffected limb of previously injured players, suggesting cross-contamination. Hierarchical ordering revealed that the eccentric phase of the CMJ contributed little to performance in previously injured players. In noninjured players, the eccentric rate of force development and concentric peak force were able to account for up to 89% of the variation in CMJ performance. ***Conclusions:*** Single-leg CMJ is advocated for player profiling, being more sensitive to previous injury, and negating the opportunity for interlimb compensation strategies. Movement strategy deficits in previously injured players suggest rehabilitation foci specific to eccentric force development.

***Keywords:*** elite male soccer, neuromuscular performance, player profiling, previous injury

The application of evidence-based practice in the context of a professional soccer club medical department relies on the clinical interpretation of player profiling, which has the potential to inform practice. Sports medicine departments utilize profiling to identify players at risk of injury and to objectively monitor the rehabilitation of injured players. The greatest risk factor for injury is previous injury, with players who have sustained a lower limb injury being up to 3 times more likely to incur a reinjury,[1] which is often of greater severity than the original injury.[2] However, the influence of injury history on neuromuscular performance in professional soccer players has received little attention.[3] Hart et al[4] recently identified interlimb asymmetries during bilateral countermovement jumps (CMJ) in professional players with a prior injury, despite no performance deficits in jump performance. Interlimb asymmetries might reflect a compensation strategy to protect previously injured systems, and they have been identified following anterior cruciate ligament injury, for example.[5,6] The bilateral strength asymmetries observed in soccer players [7–9] might reflect limb dominance, but asymmetry might also be attributable to previous injury and/or limitations in rehabilitation.[10]

The influence of previous injury on bilateral asymmetry observed using double-legged CMJ and using the healthy contralateral limb as a control negate the potential decline in the nonaffected limb as a consequence of the injury. Furthermore, the double-legged jump enables movement compensation strategies to facilitate performance outcome,[6] with interlimb asymmetry not impairing jump performance.[4] Movement strategy or how the movement is achieved is more insightful to the clinician in guiding rehabilitation than the outcome measure (eg, jump height). The potential for movement compensations[5,6] suggests that double-leg CMJ performance data should be interpreted cautiously when there is injury to a single limb. The single-leg CMJ test might therefore provide a more valid test of limb performance and interlimb asymmetry,[11] especially in a unilateral limb-dominant sport like soccer, where the primary mechanism of injury is high-speed running or cutting motions, where unilateral performance is fundamental. The opportunity for movement strategy compensations[5,6] during such activities provides little scope for bilateral compensations, and unilateral assessments might therefore be more sensitive to previous injury.

The aim of the present study was to assess the influence of previous injury in actively competing professional soccer players on bilateral and unilateral CMJ performance and movement strategy. Detailed analysis of the CMJ using force platforms has led to a richer appreciation of the neuromuscular profile that makes up the jump.[12] The present study aimed to investigate the relative sensitivity of double- and single-leg CMJ performance by creating hierarchical models of those movement strategy variables influencing performance. It is hypothesized that players with an injury history will have impaired jump performance and altered movement

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strategies. This could provide new clinical insights for the practitioner in the club setting as they decide on their choice of profiling tests and in identifying the focus for their rehabilitation prescription.

# Methods

## Design

The study was conducted as a cross-sectional design, completed as part of a broader player profiling battery conducted during the competitive season within an English Championship (tier 2) club.

## Participants

Group stratification was developed based on injury history. The "injured" group (n = 10; age = 23.1 [7.7] y, weight = 76.7 [1.4] kg, body mass index = 24.70 [2.24] kg/m$^2$, body fat = 6.9% [3.5%], professional playing history = 6.7 [7.8] y) was composed of players who were currently fit, but did have an injury history that included a significant injury to one limb only that resulted in their being unable to train with the team for ≥3 months.[13] The "uninjured" group (n = 10; age = 23.5 [2.1] y, weight = 79.8 [2.8] kg, body mass index = 24.34 [1.22] kg/m$^2$, body fat = 6.7% [0.7%], professional playing history = 6.9 [3.5] y) had never sustained a lower limb injury that resulted in >8 days lost from training and competition and were selected from the full squad (n = 36) to reflect the demographics of the "injured" group. Therefore, 20 professional soccer players from the same English Championship (tier 2) club completed the study. All participants were full-time professional outfield players with a minimum of 6 years as a professional and participating in full-time daily training at the time of testing.

All players provided written consent and were made aware that the data would remain anonymized and would not affect their standing within the team. Ethical consent was provided by the research development committee at the football club, and in accordance with the Helsinki Declaration. All players were familiar with the testing protocols as part of the regular administration of player profiling, with no difference between groups in this respect.

## Procedures

Prior to testing, all players completed an injury history form to determine their experimental grouping. All players were weighed on a Seca performance scale (model 799) and had their body fat recorded with a 7-point caliber testing procedure.[14] All testing took place between 9:30 and 10:30 AM, more than 5 days since the previous match and following a scheduled rest day. Footwear, nutritional status, and a 15-minute warm-up was standardized between trials, with 7 days separating the unilateral and bilateral jump trials, which were randomized in order. The warm-up included 3 familiarization trials performed at 50%, 75%, and 100% of maximal CMJ performance.[15,16]

The bilateral CMJ testing procedure consisted of each player performing a series of 5 jumps, with a 10-second rest between each jump. The players performed a countermovement to a self-selected depth with self-selected arm swing and were instructed to jump as fast and as high as possible.[17,18] The players were given a simple count into each jump of "3-2-1 jump" by the tester to promote reliability between trials and participants.[12]

A similar procedure was employed for the unilateral CMJ protocol. During the unilateral jump, the nontest leg was off the ground and held approximately parallel to the midshin of the jumping leg during the jump.[19] The players rested on 2 legs between trials, shifting onto the test leg when the tester started the count down and cueing of "3-2-1 jump." Successive trials were alternated between limbs, with 5 trials completed on each limb.

All experimental trials were completed on a dual force plate system (model no: PS 2141; PASPORT force plate, Pasco Roseville, CA) and subsequently analyzed using ForceDecks customized software package (NMP ForceDecks). Each force platform measured 0.35 × 0.35 m, and the vertical ground reaction force was sampled at 1000 Hz.

Initiation of the jump was defined by a 20-N change relative to passive bodyweight,[4,20] the eccentric deceleration phase from peak negative velocity to zero velocity of the mass center, and the concentric phase from zero velocity to takeoff.[5] The dependent performance variables were defined as jump height, calculated using flight time,[21] and the flight time:contraction time ratio.[12] The dependent movement strategy variables were defined as the peak force, rate of force development (RFD), duration, and impulse in the eccentric and concentric phases, the force at zero velocity, and the depth of the CMJ.

## Statistical Analyses

A univariate general linear model was defined to investigate the main effects for group (injured vs noninjured) in each of the dependent variables, for double- and single-leg CMJ trials. A preliminary analysis of the single-leg trials included both limbs from each player, with a pooled sample therefore of n = 20 in each group. Secondary analysis accounted for the specific injury history of the players in the injured group, with the affected limb considered separately to the nonaffected limb, and the dominant limb differentiating trials in the noninjured group, creating n = 10 in each group. Statistical significance was predetermined at $P \leq .05$ and supported by partial eta squared as a measure of effect size. All assumptions associated with the general linear model were investigated and, in the secondary analysis post hoc, measures were employed to investigate the main effects.

Forward stepwise regression modeling was used to determine the hierarchical ordering of force variables influencing single-leg CMJ performance, investigated using both the jump height and flight time: contraction time ratio. All force variables were entered into the equation, with strength of the correlation quantified as the correlation coefficient ($r$). Hierarchical modeling was applied to the pooled sample (n = 20) for each group, and subsequently to the reduced sample, accounting for previous history and limb specificity (n = 10).

# Results

## CMJ Performance

Figure 1 summarizes the influence of previous injury on the jump height and flight time:contraction time ratio for the double-leg and single-leg CMJ, where the injured group is also differentiated for the injured limb. There was no significant main effect for group in the double leg CMJ in jump height ($P = .447$, $\eta_p^2 = .032$) or flight time: contraction time ratio ($P = .623$, $\eta_p^2 = .014$). However, the injured group (pooled for affected and nonaffected limb) scored significantly lower in the single-leg jump height ($P = .007$, $\eta_p^2 = .176$) and flight time:contraction time ratio ($P = .005$, $\eta_p^2 = .188$) than the noninjured group. Secondary analysis revealed that the previously injured limb scored significantly lower than the noninjured players, but was not different from the nonaffected limb in jump height ($P = .014$; $P = .687$) and time ratio ($P = .017$; $P = .706$).

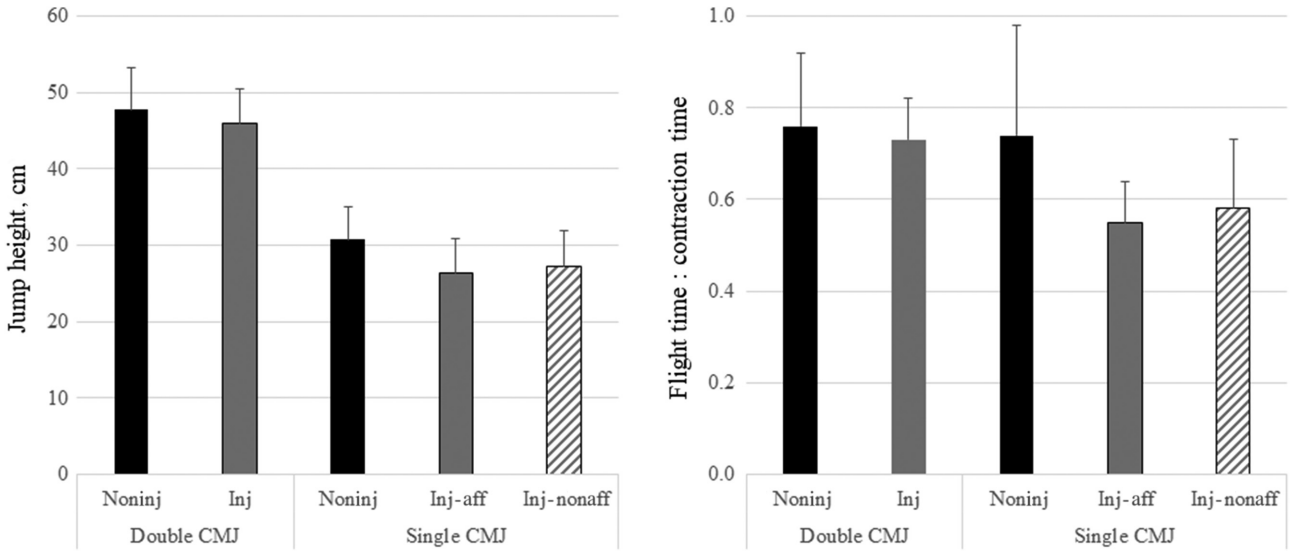


**Figure 1** — The influence of previous injury on CMJ jump height and flight time:contraction time ratio. CMJ indicates countermovement jump aff, affected; Inj, injured; nonaff, nonaffected; noninj, noninjured.

## CMJ Movement Strategy

Table 1 summarizes the influence of previous injury on the movement strategy variables for the double-leg CMJ. There was no significant difference between groups for any movement strategy variable ($P \geq .122$)

Table 2 summarizes the influence of previous injury and limb status on the movement strategy variables for the single-leg CMJ. Compared with the injured group (pooled, n = 20), the noninjured group elicited significantly ($P \leq .05$) greater peak force and RFD in the eccentric phase of the single-leg CMJ; significantly greater force at zero velocity, along with a shallower countermovement; and significantly greater peak force and RFD in the concentric phase, which was of significantly shorter duration. Further analysis revealed that these impairments in the injured group were evident in both limbs, with the exception of countermovement depth, which was significantly deeper only in the affected limb.

## Hierarchical Ordering of Factors Influencing Single-Leg CMJ Performance

Table 3 summarizes the hierarchical ordering of the movement strategy variables influencing performance of the single-leg CMJ, quantified as the square of the correlation coefficient ($r^2$) at each stage. The first listed variable has the greatest individual predictive power of performance. The model is terminated when the addition of a new movement strategy variable fails to improve the magnitude of the correlation coefficient.

In the noninjured group, concentric impulse and eccentric peak force were able to account for 44% of the variability in jump height, while eccentric RFD and concentric peak force accounted for 89% of the variability in the time ratio. In comparison, hierarchical modeling of the injured group featured only concentric phase elements. Concentric peak force and impulse accounted for 75% of the variability in jump height, while concentric phase duration and peak force accounted for 84% of the variability in the time ratio. In the injured limb, concentric peak force and RFD accounted for 70% of the variability in jump height, while concentric phase duration and countermovement depth accounted for 89% of the variability in the time ratio.

**Table 1 The Influence of Previous Injury on Double-Leg CMJ Force–Time History Metrics**

| Measure | Noninjured | Injured | *P* | $\eta_p^2$ |
|---|---|---|---|---|
| Ecc impulse, N·s | 120.29 (30.78) | 129.88 (25.53) | .290 | .029 |
| Ecc peak force, N | 807.50 (148.36) | 787.80 (108.05) | .634 | .006 |
| Ecc RFD, N·s$^{-1}$ | 2620.45 (1787.50) | 1946.91 (656.80) | .122 | .062 |
| Ecc duration, s | 0.18 (0.06) | 0.21 (0.02) | .366 | .046 |
| Force at zero velocity, N | 780.90 (149.42) | 760.90 (116.73) | .640 | .006 |
| CMJ depth, cm | 32.01 (9.64) | 37.99 (7.14) | .132 | .121 |
| Con impulse, N·s | 201.45 (34.06) | 199.27 (27.04) | .824 | .001 |
| Con peak force, N | 963.55 (111.55) | 921.69 (135.99) | .294 | .029 |
| Con RFD, N·s$^{-1}$ | 1115.90 (742.90) | 992.50 (691.90) | .590 | .008 |
| Con duration, s | 0.26 (0.04) | 0.27 (0.03) | .628 | .013 |

Abbreviations: CMJ, countermovement jump; Con, concentric phase; Ecc, eccentric phase; RFD, rate of force development.

# Discussion

The aim of the present study was to investigate the influence of previous injury on the performance and movement strategy of single- and double-leg CMJ jumps in professional soccer players, with practical implications in profiling test selection and clinical interpretation to inform rehabilitation foci. While previous research[3,4] has identified limb asymmetries in double-leg CMJ, this task is subject to interlimb compensations that might mask neuromuscular impairments in the injured limb, and soccer is largely a unilateral sport.

## Sensitivity of CMJ Performance to Previous Injury

Double-leg CMJ performance was not sensitive to previous injury, with no difference between groups and consistent with the findings

**Table 2 The Influence of Previous Injury on Single-Leg CMJ Force–Time History Metrics**

| | Noninjured | Injured (affected) | Injured (nonaffected) |
|---|---|---|---|
| Ecc impulse, N·s | 56.55 (16.82) | 65.13 (17.28) | 61.36 (13.43) |
| Ecc peak force, N | 1352.40 (177.50)** | 1252.30 (137.40) | 1236.80 (85.37)* |
| Ecc RFD, $N \cdot s^{-1}$ | 4327.30 (2838.79)** | 2335.40 (808.10)* | 2539.30 (859.00)* |
| Ecc duration, s | 0.18 (0.06) | 0.21 (0.03) | 0.19 (0.04) |
| Force at zero velocity, N | 1336.75 (180.84)** | 1213.90 (142.99)* | 1215.60 (82.56)* |
| CMJ depth, cm | 16.68 (6.83)** | 22.29 (5.75)* | 20.13 (4.68) |
| Con impulse, N·s | 170.85 (28.72) | 161.15 (18.70) | 162.46 (16.63) |
| Con peak force, N | 1917.10 (282.58)** | 1663.40 (215.06)* | 1700.80 (180.20)* |
| Con RFD, $N \cdot s^{-1}$ | 4538.95 (3050.44)** | 2257 (1331.61)* | 2595.60 (1575.08)* |
| Con duration, s | 0.28 (0.06)** | 0.33 (0.04)* | .33 (0.06)* |

Abbreviations: CMJ, countermovement jump; Con, concentric phase; Ecc, eccentric phase; RFD, rate of force development.
*Significantly different from noninjured ($P \leq .05$). **Noninjured significantly ($P \leq .05$) different from injured (pooled n = 20).

**Table 3 Hierarchical Ordering of Factors Influencing Single-Leg CMJ Performance**

| | Step 1 | Step 2 |
|---|---|---|
| | Noninjured (n = 20) | |
| Jump height | Con impulse ($r^2 = .35$) | Con PkForce ($r^2 = .89$) |
| FT:CT ratio | Ecc RFD ($r^2 = .78$) | Ecc PkForce ($r^2 = .44$) |
| | Injured (n = 20) | |
| Jump height | Con PkForce ($r^2 = .38$) | Con impulse ($r^2 = .75$) |
| FT:CT ratio | Con duration ($r^2 = .77$) | Con PkForce ($r^2 = .84$) |
| | Injured and affected (n = 10) | |
| Jump height | Con PkForce ($r^2 = .47$) | Con RFD ($r^2 = .70$) |
| FT:CT ratio | Con duration ($r^2 = .71$) | CMJ depth ($r^2 = .89$) |

Abbreviation: CT, contraction time; CMJ, countermovement jump; Con, concentric phase; Ecc, eccentric phase; FT, flight time; PkForce, peak force; RFD, rate of force development.

of Hart et al.[4] This comparable performance might be attributed to interlimb compensations, since the injured group also displayed no difference in the magnitude of the movement strategy variables in the double-leg CMJ. Force–time history metrics in both the eccentric and concentric phases, along with CMJ depth, were consistent between groups. In contrast, single-leg CMJ, which does not allow for movement compensations in the contralateral limb, was sensitive to previous injury, in respect to performance outcomes and movement strategy. The injured group (pooled for affected and nonaffected limbs) produced significantly impaired performance compared with the noninjured group in both jump height and flight time:contraction time ratio. Single-leg CMJ is therefore advocated clinically, being sensitive to previous injury, negating the opportunity for a player to mask injury via interlimb compensations, and providing greater functional specificity to the common mechanisms of injury.

Of note, the nonaffected and affected limb in the previously injured players produced lower performance relative to the noninjured players, highlighting impaired bilateral asymmetry. This might be attributed to cross-contamination of the contralateral limb as a result of physical deconditioning in the aftermath of the original injury, reflecting previous observations in bilateral movement compensations in previously injured players.[4–6] Additional contributing factors might include a lack of specific rehabilitation targeting movement strategy or early cessation of rehabilitation,[10] or, anecdotally from conversations with players, a perceived lack of capacity by the player and a reluctance to expose themselves to previously challenging movements that might be associated with pain or risk of injury. All players were pain free and competing with no daily pain, so any residuals in the results were true residuals rather than pain-related abnormalities and deficits. Basing clinical decisions on performance outcomes should therefore be treated with caution, since movement strategy must be recovered in addition to performance outcome.

## Sensitivity of CMJ Movement Strategy to Previous Injury

Previous injury was observed to significantly influence movement strategy in a range of force–time history metrics. The noninjured players elicited greater peak force and RFD in the concentric and eccentric phases of the jump. Phase duration was also lower in the noninjured players, significantly so in the concentric phase of the jump. The noninjured players produced greater force at the point of zero velocity, and a shallower countermovement depth. These movement strategy differences highlight the range of neuromuscular compromises evident in the players with an injury history, and it is these variables that offer the practitioner the greatest opportunity for enhancing rehabilitation and ultimately reducing the risk of reinjury. These technical factors are modifiable,[22] and thus, with appropriately targeted rehabilitation and exercise prescription, the movement strategy can be regained. Since the specific injury type, location, and mechanism were not included in the group stratification, it is not possible to expand the interpretation of how specific injury mechanisms impacted upon movement strategy. With an appropriate data set, this would be an interesting opportunity for future research.

Previous injury was observed to impair the concentric and eccentric phases of the single-leg CMJ. Hierarchical ordering of the factors influencing single-leg CMJ performance highlighted that noninjured players had primary predictors from eccentric and concentric force–time history metrics. However, players with an injury history had primary influencing factors only in the concentric phase of the jump. In the affected limb, the 2 primary predictors of performance from the concentric phase were able to account for

70% of the variability in jump height and 89% of the variability in flight time:contraction time ratio. This suggests that the eccentric phase is making very little contribution to task outcome in those players who have suffered a previous injury. This has clear practical implications in rehabilitation, as eccentric muscular actions are commonly cited in common mechanisms of injury.[23–25] Deceleration prior to landing and cutting is a common mechanism of anterior cruciate and ankle ligamentous injury, for example.[26,27] Noting the impaired rate of eccentric force development in players with an injury history, eccentric contraction of the hamstrings in high-speed running is commonly cited as being the primary mechanism of hamstring strain injury, and eccentric hamstring strength is commonly cited as the primary modifiable aetiological factor.[28] Of note, these injures also suffer from high reinjury rates,[1] further highlighting the need for effective rehabilitation. These movements are also fundamental to the intermittent, multidirectional, and unilateral demands of soccer. Practically, the failure of previously injured players to maximize the eccentric phase to develop performance outcomes can inform a rehabilitative focus in eccentric strength and RFD in both limbs and monitored using the single-leg CMJ. While double-leg CMJ tests might still have value in investigating interlimb asymmetries within a bilateral task, it should be acknowledged that bilateral tasks are relatively uncommon in the technical and physical demands of the game, or in the common mechanisms of injury. Where concerns have been raised about single-leg CMJ testing with respect to the technical complexity of the test,[10] in a cohort of professional soccer players, this does not present an issue.

### Generalizability of Findings

The current study was conducted within an English Championship professional soccer club, providing high ecological validity relative to the epidemiology literature typically derived from elite male soccer. However, generalizations beyond this specific cohort should be treated with caution and the sample size is inevitably limited by squad size.[29] Furthermore, the classification of "injured" players while retaining a sample of 10 did not consider specific injury types or playing position, or the time frame of the injury in relation to the testing. It would be interesting, for example, to investigate the specific influence of hamstring strain injury and the duration of influence of the previous injury, given the relatively high incidence and recurrence rates and the association with eccentric contractile properties. This would, however, require data to be pooled across clubs or a longitudinal study within the same club, both paradigms presenting a myriad of confounding variables. Performance of the players with an injury history might also be compared against baseline preinjury data, as opposed to noninjured players. In this respect, baseline screening of single-leg CMJ performance and movement strategy will inform practice within our club. Testing and research can be challenging in a professional sporting environment, and the testing in the present study was specifically designed to be player friendly. Integrating CMJ testing more frequently within the players' schedule might also enhance familiarity, which has been shown to decrease interlimb asymmetries.[30]

## Conclusions

Double-leg CMJ performance and movement strategy were not sensitive to previous injury in a cohort of professional soccer players, with interlimb compensation strategies masking the influence of neuromuscular deficits and reflecting the interlimb asymmetries highlighted in previous studies.[3,4] The single-leg CMJ was sensitive to previous injury, both in performance outcome measures and movement strategy. Single-leg CMJ testing is therefore advocated as a more efficacious test, sensitive to previous injury and more functionally specific to the physical demands and common injury mechanisms in soccer. Movement strategy deficits were observed in the concentric and eccentric phases of the single-leg CMJ, with implications for rehabilitation foci, but hierarchical ordering of factors influencing performance highlighted a lack of eccentric phase contribution in players with an injury history. It is suggested that the movement strategy, rather than the movement outcome, should be the priority in an evidence-based rehabilitation plan. Compromised or even compensatory movement strategies warrant clinical attention, even if they result in gross performance outcomes comparable with healthy baseline or normative values. Deficits in the nonaffected limb of previously injured players also suggest some cross-contamination of the contralateral limb, which should further inform rehabilitation.

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## References

1. Hagglund M, Walden M, Ekstrand J. Previous injury as a risk factor for injury in elite football: a prospective study over two consecutive seasons. *Br J Sports Med*. 2006;40(9):767–772. PubMed ID: 16855067 doi:10.1136/bjsm.2006.026609
2. Ekstrand J, Krutsch W, Spreco A, et al. Time before return to play for the most common injuries in professional football: a 16-year follow-up of the UEFA Elite Club Injury Study. *Br J Sports Med*. 2020; 54(7):421–426. PubMed ID: 31182429 doi:10.1136/bjsports-2019-100666
3. Cohen D, Clarke N, Harland S, et al. Are force asymmetries measured in jump tests associated with previous injury in professional footballers? *Br J Sports Med*. 2014;48(7):579–580. doi:10.1136/bjsports-2014-093494.53
4. Hart LM, Cohen D, Patterson SD, et al. Previous injury is associated with heightened countermovement jump force-time asymmetries in professional soccer players even after return to play. *Trans Sports Med*. 2019;2(5):256–262. doi:10.1002/tsm2.92
5. Jordan M, Aagaard P, Herzog W. Lower limb asymmetry in mechanical muscle function: a comparison between ski racers with and without ACL reconstruction. *Scand J Med Sci Sports*. 2015;25(3): e301–e309. PubMed ID: 25212216 doi:10.1111/sms.12314
6. Baumgart C, Schubert M, Hoppe MW, et al. Do ground reaction forces during unilateral and bilateral movements exhibit compensation strategies following ACL reconstruction? *Knee Surg Sports Traumatol Arthrosc*. 2017;25(5):1385–1394. PubMed ID: 25957607 doi:10.1007/s00167-015-3623-7
7. Croisier JL, Reveillon V, Ferret JM, et al. Isokinetic assessment of knee flexors and extensors in professional soccer players. *Isokinet Exerc Sci*. 2003;11(1):61–62. doi:10.3233/IES-2003-0124
8. Rahnama N, Lees A, Bambaecichi E. A comparison of muscle strength and flexibility between the preferred and non-preferred

leg in English soccer players. *Ergonomics*. 2005;48(11–14):1568–1575. PubMed ID: 16338722 doi:10.1080/00140130500101585
9. Fousekis K, Tsepis E, Vagenas G. Multivariate isokinetic strength asymmetries of the knee and ankle in professional soccer players. *J Sports Med Phys Fitness*. 2010;50(4):465–474. PubMed ID: 21178934
10. Impellizzeri FM, Rampinini E, Maffiuletti N, et al. A vertical jump force test for assessing bilateral strength asymmetry in athletes. *Med Sci Sports Exerc*. 2007;39(11):2044. PubMed ID: 17986914 doi:10.1249/mss.0b013e31814fb55c
11. Benjanuvatra N, Lay BS, Alderson JA, et al. Comparison of ground reaction force asymmetry in one- and two-legged countermovement jumps. *J Strength Cond Res*. 2013;27(10):2700–2707. PubMed ID: 23287834 doi:10.1519/JSC.0b013e318280d28e
12. McMahon JJ, Suchomel TJ, Lake JP, et al. Understanding the key phases of the countermovement jump force time curve. *Strength Cond J*. 2018;40:96–106. doi:10.1519/ssc.0000000000000375
13. King DA, Hume PA, Milburn PD, et al. Match and training injuries in Rugby League. *Sports Med*. 2010;40(2):163–178. PubMed ID: 20092367 doi:10.2165/11319740-000000000-00000
14. Gabbett T, Georgieff B, Domrow N. The use of physiological, anthropometric, and skills selection in a talent-identified junior volleyball squad. *J Sports Sci*. 2007;25(12):1337–1344. PubMed ID: 17786686 doi:10.1080/02640410601188777
15. Byrne PJ, Moody JA, Cooper SM, et al. The reliability of countermovement jump performance and the reactive strength index in identifying drop-jump drop height in hurling players. *OA J Exerc Sports Med*. 2017;1(1):1–10.
16. Sole CJ, Mizuguchi S, Sato K, et al. Phase characteristics of the countermovement jump force-time curve: a comparison of athletes by jumping ability. *J Strength Cond Res*. 2017;32(4):1155–1165. doi:10.1519/JSC.0000000000001945
17. Laffaye G, Wagner PP, Tombleson TIL. Countermovement jump height: gender and sport-specific differences in the force-time variables. *J Strength Cond Res*. 2014;28(4):1096–1105. PubMed ID: 23838969 doi:10.1519/JSC.0b013e3182a1db03
18. McLellan CP, Lovell DI, Gass GC. The role of rate of force development on vertical jump performance. *J Strength Cond Res*. 2011;25(2):379–385. PubMed ID: 20093963 doi:10.1519/JSC.0b013e3181be305c
19. Bromley T, Turner AN, Read P, et al. The effects of a competitive soccer match on jump performance and inter-limb asymmetries in elite academy soccer players. *J Strength Cond Res*. 2018. doi:10.1519/jsc.0000000000002951
20. Heishman AD, Daub BD, Miller RM, et al. Countermovement jump reliability performance with and without an arm swing in NCAA division 1 intercollegiate basketball players. *J Strength Cond Res*. 2020;34(2):546–558. doi:10.1519/JSC.0000000000002812
21. Linthorpe NP. Analysis of standing vertical jumps using a force platform. *Am J Physics*. 2001;69:1198–1204. doi:10.1119/1.1397460
22. Jordan M, Aagaard P, Herzog WA. Comparison of lower limb stiffness and mechanical muscle function in ACL-reconstructed, elite, and adolescent alpine ski racers/ski cross athletes. *J Sport Health Sci*. 2018;7(4):416–424. PubMed ID: 30450249 doi:10.1016/j.jshs.2018.09.006
23. Kirkendall DT, Garrett WE. Clinical perspectives regarding eccentric muscle injury. *Clin Orthop Related Res*. 2002;403:S81–S89. doi:10.1097/00003086-200210001-00010
24. La Stayo PC, Woolf JM, Lewek MD, et al. Eccentric muscle contractions: their contribution to injury, prevention, rehabilitation, and sport. *J Orthop Sports Phys Ther*. 2003;33(10):557–571. doi:10.2519/jospt.2003.33.10.557
25. Thelen D, Chumanov ES, Sherry MA, et al. Neuromusculoskeletal models provide insights into the mechanisms and rehabilitation of hamstring strains. *Exerc Sport Sci Rev*. 2006;34(3):135–141. PubMed ID: 16829741 doi:10.1249/00003677-200607000-00008
26. Boden BP, Dean GS, Feagin JA, et al. Mechanism of anterior cruciate ligament injury. *Orthop*. 2000;23(6):573–578. doi:10.3928/0147-7447-20000601-15
27. Fong DTP, Hong Y, Chan LK, et al. A systematic review on ankle injury and ankle sprain in sports. *Sports Med*. 2007;37(1):73–94. PubMed ID: 17190537 doi:10.2165/00007256-200737010-00006
28. Ekstrand J, Walden M, Hagglund M. Hamstring injuries have increased 4% annually in men's professional football, since 2001: a 13-year longitudinal analysis of the UEFA Elite Club injury study. *Br J Sports Med*. 2016;50(12):731–737. doi:10.1136/bjsports-2015-095359
29. Lonergan B, Senington B, Patterson SD, et al. The effect of fatigue on phase specific countermovement jump asymmetries in ACL-R and non-injured rugby union players. *Trans Sports Med*. 2018;1(6):238–249. doi:10.1002/tsm2.48
30. Bishop C, Turner A, Read P. Effects of inter-limb asymmetries on physical and sports performance: a systematic review. *J Sports Sci*. 2018;36(10):1135–1144. PubMed ID: 28767317 doi:10.1080/02640414.2017.1361894

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